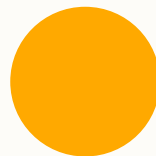




Javascript Conditionals

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Javascript

- Let's take a look at the structure of the code we have written so far
- Linear
- Everything exists at the "top level"
- Things will run always
 - Unless we commented them with //





Javascript

- This has suited us fine so far
- But if we wanted to write larger program
- We would need some way of controlling the flow of our code





Javascript

- Manifests as
 - "Run this code sometimes, and this other code a different time"
- *If this, do that*
- Let's consider a few cases where this problem might arise





Javascript

- If user height is greater than 6 foot
 - Alert a special message that they are tall
- If the user is shorter than a giraffe
 - Alert a message telling the user
- We need new JS syntax for this





Javascript

- *if* statement
 - Conditional block
 - Expression block
 - *if (conditional){
 expression;
}*
- 



Javascript

- *Conditional*
 - Boolean (true or false type)
- Often defined
 - ==
 - >
 - <
- == is not the same as =
- Can negate with a !





Javascript

- These will evaluate to either (true or false)

var height = 15;

var isTall = height > 20; // false

var isSixFoot = height == 6; // false

var isShort = !isTall; // true





Javascript

- Aside: String equality
- JS has some quirks
 - == does not work as expected on strings
 - Need to use ===
- Reference vs value equality





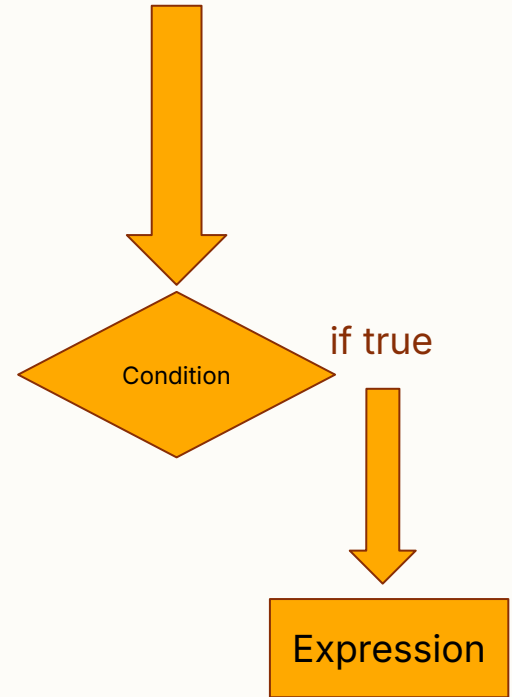
Javascript

- Expressions
 - Just any block of code*
 - Everything you've written so far has been a set of expressions
- But now we have everything we need for



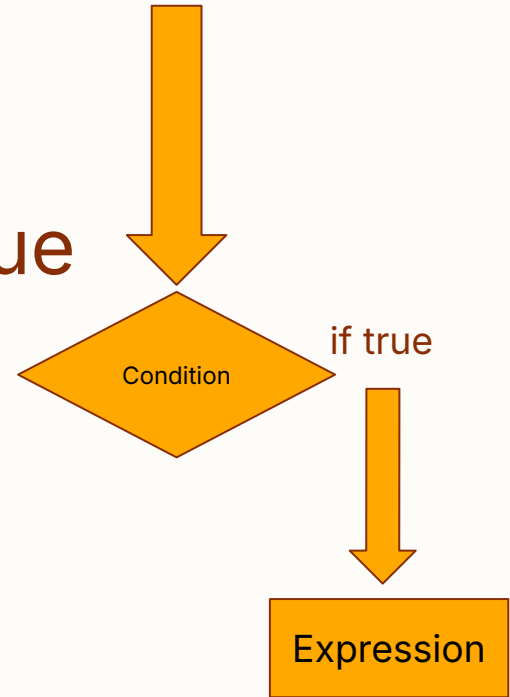
Javascript

- `if(condition){
 expression_if_true
}`
- `if (height > 5){
 alert("tall!")
}`



Javascript

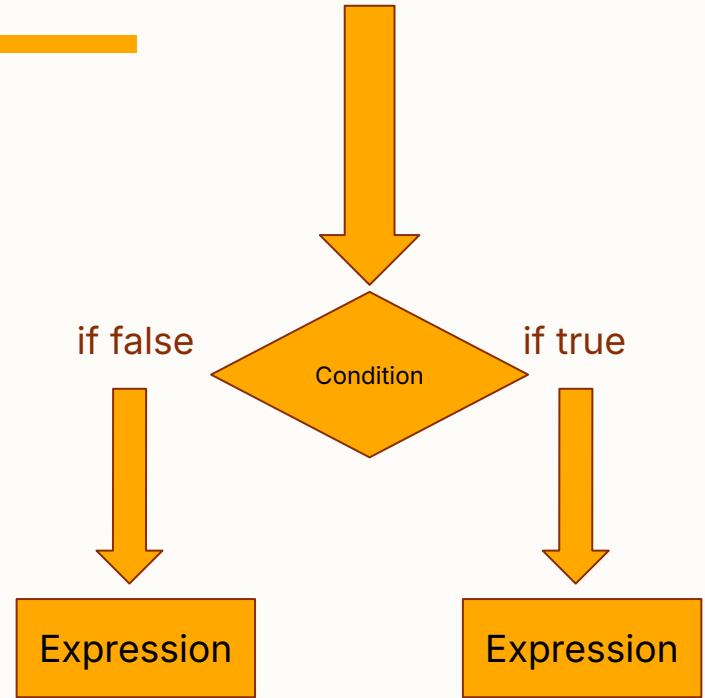
- Execute the inner block
 - Only if the condition == true
- But we can do more
- Would be nice if we had the opposite



Javascript

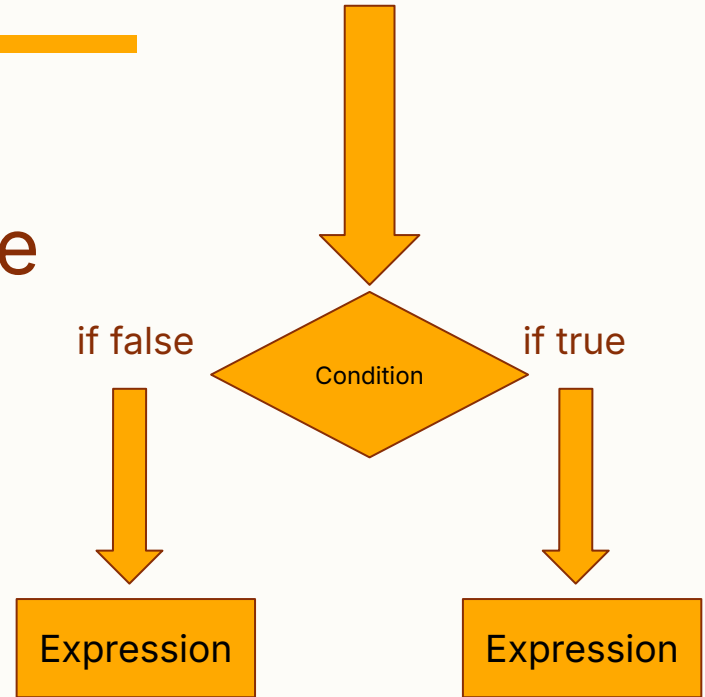
- Else block

```
if(conditional){  
    expr_when_true  
}  
else {  
    expr_when_false  
}
```



Javascript

- Important note
- A condition is either true
 - Or it is false
- Has some implications
 - One of those expr. must therefore run



Javascript

- Can check for multiple conditionals
- Using an *else if* block

```
if (conditional_1){  
    Expr_1;  
} else if (conditional_2){  
    Expr_2;  
} else if(conditional_3){  
    Expr_3;  
} else {  
    Expr_4;  
}
```

- Else here will act as a fallback
- If none of the conditionals were ever true
- Then we will eventually hit the else statement

Javascript

- Let's practice this a bit
- Grade calculator
- Given a percentage
- We can play around with values
- Special cases?
- Modify a "global" var

